

Yu-Ting (Harry) Liao

designer & computational artist — the body as an interface

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PROFILE

I design the **body as an interface** — wearable instruments, embodied sound, and real-time AI systems that let a person extend, feel, and learn through their own body. My path runs from fashion pattern-cutting (UAL) through generative-AI media to computational arts, grounded in professional visual-design practice in Taipei (2020–25). Current focus: **Solo Choir**, a wearable instrument that turns one singer into a choir.

EDUCATION

2025 — 26 **MA Computational Arts** — Goldsmiths, University of London (expected Sep 2026)
Physical computing, embodied interaction, real-time audio-visual systems. Thesis: **Solo Choir** (below).

2022 — 24 **MA Digital Multimedia Arts** — Shih Hsin University, Taipei
Thesis: generative-AI NPCs — a GPT-4 dialogue system in Unreal Engine 5 integrating speech recognition, synthesis, and sentiment analysis for real-time, context-aware interaction.

2016 — 19 **BA Fashion Pattern Cutting** — University of the Arts London

SELECTED PROJECTS

2026 **Solo Choir** — MA thesis · wearable instrument
Turns one singer into a choir: diatonic SATB harmony re-synthesised in the singer's **own voice**, delivered into the body via open-ear bone conduction (learning) and performed from it as a simultaneous chord (surface exciters). Real-time neural voice conversion on CPU, ~47 ms. Research question: does sending harmony *into / from* the body change how singers learn and perform it, versus harmony mediated by screens or speakers? [case study →](#)
neural voice conversion · real-time DSP · bone conduction · hardware prototyping

2026 **Choir of Three (and One)** — sound / AI
A DDSP model trained on ~10 minutes of my own singing performs three parts of a Bach chorale; the AI yields to silence when I hold my own part — a study in mutual yielding as the core of choral practice.
DDSP · Tone.js · ml5.js · p5.js

2026 **Fabri-Form** — robotics
A robotic interface for pattern-cutting: a multimodal LLM reads a hand-drawn sketch, generates 2D cutting paths, and a Pico-driven laser traces them onto fabric.
multimodal LLM · Raspberry Pi Pico · laser

2025 **The Bonnet** — wearable
An anti-surveillance headpiece: a custom-trained YOLOv8 model spots a CCTV camera and a servo drops a fabric veil over the face — reclaiming the gaze.
YOLOv8 · OpenCV · Arduino

2025	Gesture-Float — hardware A wireless flying unit steered by the arm: an arm-mounted Pico W streams hand pitch and roll over Wi-Fi to a helium balloon unit, hand-soldered for stable, organic flight. Pico W · UDP · PCB · soldering
2025	Draping Duet — performance A real-time audiovisual installation reframing fashion draping as choreography — StreamDiffusion turns the atelier into flowing silk while a microphone amplifies the friction of the cloth. StreamDiffusion · TouchDesigner · OBS
EXHIBITIONS	
2026	Pop-Up · Harmonizing! — PSH Atrium, Goldsmiths, London Exhibited a live vocal-harmony instrument; also designed the show's graphics and caption cards for all 37 works.
EXPERIENCE	
2024 — 25	Visual Designer — Nice Crush Design House, Taipei Digital advertising, social graphics, packaging; cohesive visual communication across online and offline campaigns.
2021 — 22	Visual Designer — Digibionic, Taipei Brand-marketing visual assets; marketing materials, promotions, and product logistics.
2020 — 21	Graphic Designer — AY-AY International, Taipei Graphic assets and layouts across multimedia projects; produced and maintained style guides.
SKILLS	
physical computing	Raspberry Pi Pico W, Arduino, servo/motor control, circuit soldering, sensor fusion
creative code	p5.js, Tone.js, ml5.js, TouchDesigner, Web Audio API, JavaScript
machine learning	YOLOv8, OpenCV, Python, DDSP, neural voice conversion, dataset curation (Roboflow)
3d / game engine	Unreal Engine 5, Unity
design	Adobe Illustrator, Photoshop, InDesign, After Effects; pattern cutting & garment construction
references and full documentation available on request · yvwhy.github.io	